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BSC (HONS) COMPUTER SCIENCE (CYBER SECURITY)

MAL3018 COMPUTING PROJECT

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1.0 Project Title
Automated IT Security Audit System for Windows Endpoints

2.0 Keywords
Automated Security Audit, Windows Endpoint Security, Portable Audit Tool, Security Configuration, IT Asset Inventory, Risk Prioritization, Security Reporting, Centralized Dashboard

3.0 Links
GitHub:
https://github.com/lowyanhui04-cell/EndpointLens.git
PebblePad:
https://v3.pebblepad.co.uk/spa/#/public/mtw79ZnbtfpZbxWkWy3brbbmrW

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6.0 Background and Problem Statement
Organizations perform IT audits to understand their computer assets, review security controls, and document weaknesses that require attention. For small and medium-sized organizations with limited IT resources, a manageable audit process can support practical cybersecurity risk management. The National Institute of Standards and Technology (NIST) small business guide provides a framework for organizations beginning or improving this process (Locascio, 2024).

In some small and medium-sized organizations, selected IT audit activities are performed manually. IT personnel remotely access individual computers or visit the premises to collect computer specifications, inspect installed applications and user information, check antivirus and firewall status, and record findings. The prevalence and effort of this workflow will be investigated during requirements gathering rather than assumed to apply to every organization.

This manual process can require considerable administrative effort when several Windows computers must be inspected. Findings recorded in separate documents or spreadsheets can make it difficult to compare endpoints and identify the latest

available evidence. Differences in checking procedures and manual transcription may also produce inconsistent records or omissions.

Existing endpoint management and security platforms already automate many relevant activities, including device management, security monitoring, configuration assessment, and threat detection. The project therefore does not claim that automated auditing is new. Its proposed contribution is a focused implementation of selected configuration checks with visible evidence, understandable recommendations, and a small-scale centralized reporting workflow. Whether this approach is sufficiently useful and lightweight will be assessed through testing.

The proposed system will collect computer specifications, installed application inventory, current interactive user information, local account information, antivirus status, firewall status, and a defined set of security configuration values. A rule-based engine will evaluate this evidence and produce audit results, severity levels, remediation recommendations, and structured reports. Backup inspection is excluded. Each audit reflects the endpoint state at the time of collection; the dashboard shows timestamps and does not imply that historical information is continuously current.

The primary users are IT administrators, IT support personnel, and internal auditors responsible for small Windows endpoint environments. They need to identify basic configuration weaknesses and prepare consistent reports. The system supports their assessment and follow-up decisions; it does not replace professional judgment, antivirus protection, or a complete organizational compliance audit.

7.0 Objectives

The project aims to develop an automated IT security audit system with the following objectives:

- To develop a Windows endpoint agent that automatically collects at least six categories of IT asset and security information, including computer specifications, installed applications, current users, antivirus status, firewall status, and selected security configurations.
- To develop a rule-based audit engine that evaluates the collected endpoint information against at least 10 predefined IT and security audit rules and automatically identifies non-compliant or potentially insecure configurations.
- To implement a risk assessment and recommendation mechanism that classifies identified audit findings into at least three severity levels and provides corresponding security recommendations for addressing identified weaknesses.
- To develop a centralized dashboard and automated reporting module that presents endpoint information, audit findings, risk levels, and security recommendations, while generating structured audit reports for multiple Windows endpoints.

8.0 Scope of the Project

8.1 In Scope

- Windows 10 and Windows 11 desktop or laptop endpoints in a controlled, authorized test environment. Record OS edition and build; testing Windows 10 compatibility does not establish that a build is supported or secure.
- A portable .NET agent with a simple desktop interface, administrator elevation where required, and read-only collection using PowerShell and Windows interfaces.
- Seven evidence categories and 10 predefined baseline checks. Application inventory is restricted to enumerated sources, such as uninstall registry entries; it is not guaranteed to include every portable application.
- An authenticated web dashboard, central storage, endpoint audit history, severity filtering, HTML reports, and CSV or JSON export.
- On-demand audit execution with authenticated HTTPS submission. Local JSON export supports retry or manual import when submission is unavailable.
- Testing on at least three endpoints or virtual machines, including different configurations and repeated audit runs.

8.2 Out of Scope

- Backup status assessment, backup recovery testing, and backup monitoring.
- macOS, Linux, Windows Server, domain-wide deployment, and Active Directory policy assessment.
- Continuous real-time monitoring, automatic scheduled audits, email alerts, automatic remediation, remote command execution, and device isolation.
- Malware detection, penetration testing, CVE matching, SIEM integration, EDR or XDR functionality, and full CIS or NIST certification.
- Passwords, document contents, browser history, private keys, and employee activity tracking. The current user field is a session snapshot and does not prove device ownership.

9.0 Literature Review / Related Works

9.1 Overview

Several types of existing tools can support parts of an IT security audit for Windows computers. They can be grouped into four categories:

1) Endpoint management platforms such as Microsoft Intune,
2) Configuration assessment tools such as CIS-CAT,
3) Vulnerability scanners such as Tenable Nessus, and
4) Endpoint protection products such as Avira and Microsoft Defender for Endpoint.

Published guidance from the Center for Internet Security (CIS) and the National Institute of Standards and Technology (NIST) also describes what a good audit

should check. This section reviews each category, identifies its strengths and weaknesses for a small organization, and explains how the proposed system is positioned.

9.2 Manual Audit using Checklists

In the manual approach, IT personnel log in to each computer, open settings such as System Information, Programs and Features, Windows Security, and Windows Defender Firewall, and record the results in a document or spreadsheet. The method needs no software budget and is easy to start. However, it is slow when many computers must be checked, depends on the skill and care of the person performing it, and can produce records in different formats. NIST SP 800-115 notes that manual review techniques are time-consuming and that automated tools can make technical assessment more consistent (Scarfone, 2008). This weakness is the project's main motivation.

9.3 Review of Existing Tools

This section reviews the four categories of existing tools identified in Section 9.1 and evaluates each one against the needs of a small organization.

(a) Endpoint management platform: Microsoft Intune

Microsoft Intune is a cloud-based endpoint management service. It allows organizations to enrol devices, deploy configuration profiles and security baselines, define compliance policies such as requiring a firewall or antivirus, and view hardware and application inventory for managed devices (Microsoft, 2026a).

Strengths: Intune offers broad, continuous management of many devices, integrates with Microsoft Entra ID and Microsoft Defender, and can enforce settings automatically rather than only reporting them.

Weaknesses: Intune is licensed per user through a subscription (Microsoft, 2026b). Each device must be enrolled and connected to the Microsoft cloud, which requires planning and administrative knowledge. This is because it is designed for ongoing management, it is less suitable when an IT auditor or support technician only needs to perform a one-time assessment of computers that are not enrolled, such as during an audit for a client organization. Its compliance view mainly shows whether a device is compliant with a policy, rather than producing a self-contained audit report that explains each finding with evidence.

(b) Configuration assessment tool: CIS-CAT

The CIS Configuration Assessment Tool (CIS-CAT) compares a computer’s settings with the CIS Benchmarks, which are consensus-based secure configuration guides. CIS-CAT Lite is free but supports only a limited set of benchmarks (such as Microsoft Windows 10, Google Chrome, and Ubuntu) and HTML reports, while CIS-CAT Pro

supports all CIS Benchmarks, remote assessment, and the CIS-CAT Pro Dashboard and is provided through a paid CIS SecureSuite membership (CIS-CAT, 2026).

Strengths: CIS-CAT is based on an authoritative and widely recognized standard. It provides very detailed checks and a score that shows how closely a computer follows the benchmark.

Weaknesses: A full Windows benchmark contains hundreds of recommendations, and the results are written using benchmark terminology. This level of detail is useful for security specialists but can be difficult for general IT support staff to interpret and prioritize. The free version does not provide a combined view of several computers, and the tool focuses on configuration settings rather than an IT asset inventory such as hardware specifications, installed applications, and current users.

(c) Vulnerability scanner: Tenable Nessus

Nessus is a widely used vulnerability scanner. It uses plugins to detect known vulnerabilities, missing patches, and misconfigurations on hosts in a network. Nessus Essentials is free for non-commercial use but limited to five IP addresses, while Nessus Professional is a paid product that adds features such as configuration and compliance audits (GeeksforGeeks, 2026).

Strengths: Nessus has a very large vulnerability database and can scan many computers from one place. It is effective for finding known software weaknesses (CVEs) that attackers may exploit.

Weaknesses: Accurate Windows results usually require a credentialed scan, which needs administrator credentials and suitable network and firewall configuration on each target. Reports can contain a large number of technical findings, and compliance auditing requires a paid licence. Its main purpose is vulnerability discovery, not producing an IT audit record that combines asset inventory, user information, and basic security configuration in one report. Vulnerability scanning is therefore complementary to, and outside the scope of, this project.

(d) Endpoint protection: Avira and Microsoft Defender for Endpoint

Avira Free Security is an antivirus product for home and small office users that provides real-time malware protection and extra tools such as a software updater (Avira, 2026). Microsoft Defender for Endpoint is an enterprise platform with prevention, detection, investigation, and response capabilities (Mirolyubov et al., 2025).

Strengths: These products actively protect a computer from threats, which an audit tool does not do. Defender for Endpoint also provides strong visibility for large organizations.

Weaknesses: Avira protects the single computer on which it is installed and is not designed to audit settings such as local accounts or firewall profiles across several computers or to produce an audit report. Defender for Endpoint requires enterprise

licensing and onboarding and is broader than what a small organization may need for a basic audit. The proposed system does not replace these products; instead, it reads the antivirus status reported by Windows to confirm whether protection is present and enabled.

9.4 Audit Guidance and Frameworks

CIS Control 1 emphasizes maintaining visibility of enterprise assets, while CIS Control 2 addresses software inventory and management. These principles motivate the hardware and application inventory modules. The prototype will collect a limited inventory from Windows endpoints on which the agent is run; it will not discover every network asset or enforce application authorization.

CIS Control 4 calls for maintaining secure configurations of assets and software, and NIST SP 800-128 explains that configuration settings should be monitored against an approved baseline (Johnson et al., 2011). This supports examining firewall, account, and related configuration evidence. The project will document each rule and its rationale so that a project baseline is clearly distinguished from a formal benchmark assessment.

The NIST CSF 2.0 Small Business Quick-Start Guide supports organizations with modest cybersecurity resources (Locascio, 2024). It provides context for presenting findings in a form that helps IT personnel prioritize work. It does not prescribe the numeric score used in this project, and a configuration score cannot demonstrate compliance with the entire framework.

9.5 Comparison of Existing Systems

Table 1 compares selected representative tools from the categories reviewed above with the proposed system, focusing on their suitability for a small Windows environment. Avira is used as the representative endpoint protection product in the comparison.

Criteria	Microsoft Intune	CIS-CAT	Nessus	Avira	Proposed System
Main Purpose	Manage and enforce device settings	Check settings against CIS Benchmarks	Find known vulnerabilities and missing patches	Protect against malware	Audit inventory and basic security settings, then report
Cost for a Small Organization	Paid subscription per user	Lite free (limited); Pro needs paid	Essentials free (16 devices), Professional	Free version, paid Pro	No license cost (Open-Source Tools)

		membership	paid		
Setup Needed	Enrol every device in the cloud	Run assessor on each device	Install scanner; admin credentials for deep scans	Install on each device	Portable agent, run on demand
Hardware & Software Inventory	Yes	Not the main focus	Partial	No	Yes (6 Evidence Categories)
Security Configuration Check	Yes (Policies)	Yes, very detailed	Paid editions	Own protection status only	Yes, 10 focused rules
Easy-to-Understand Findings	Compliant / Not Compliant	Technical benchmark wording	Technical plugin output	Simple alerts	Evidence + Severity + Recommendation
Multi-Computer View	Yes (Cloud Portal)	Pro only	Yes	No (Free Version)	Yes (Local Dashboard)
Main Limitation	Cost and setup effort	Complex for non-specialists	Cost and technical output	Not an audit tool	Fewer checks; Windows 10/11 only

Table 1: Comparison of Existing Systems and the Proposed System

9.6 Research Gap and Contribution of the Proposed System

The review shows that automated endpoint auditing is not new, and each existing tool is strong in its own area. However, the reviewed tools have different primary purposes: Intune focuses on endpoint management, CIS-CAT on detailed benchmark compliance, and Nessus on vulnerability assessment, while endpoint protection products such as Avira and Microsoft Defender for Endpoint focus primarily on protecting endpoints and detecting security threats. For a small organization, an internal auditor or an IT support technician who needs a quick and consistent audit of several Windows computers, these tools may be too costly, too complex to set up, or produce output that is difficult to interpret.

The proposed system aims to fill this gap by combining the following in one lightweight workflow:

- **Portable and on-demand:** the agent runs without permanent installation or cloud enrolment.
- **Inventory and security in one report:** hardware, applications, users, accounts, antivirus, firewall and selected configuration settings are collected together.
- **Understandable results:** each finding shows the evidence, a severity level, and a plain-language recommendation.
- **Transparent rules:** every rule is documented and versioned, and missing data is reported as UNKNOWN instead of PASS.
- **Low cost and local:** it uses free technologies and a local server, with no subscription.
- **Centralized comparison:** a small dashboard compares several endpoints and their audit history with timestamps.

The project does not claim to provide better protection than these products. It provides a focused, transparent collection-to-report process for small test environments. Evaluation will compare it against a manual checklist using the same checks, rather than assuming that a smaller scope automatically produces greater usability or better security.

10.0 Methodology

10.1 Project Approach / Framework

An iterative prototyping approach will be used. Requirements and rule definitions will be agreed first, followed by short implementation and evaluation cycles. Early demonstrations will identify collection gaps and confusing outputs before the final integrated system is completed. The agent will be implemented before the central dashboard so that unreliable endpoint evidence is addressed at its source.

10.2 Tools and Technologies

Component	Proposed technology	Purpose
Endpoint Agent	C# and .NET desktop GUI	Portable executable and audit controls
Collectors	PowerShell and Windows CIM or registry interfaces	Read-only local evidence collection
Central Service	ASP.NET Core Web API	Validate submissions and execute versioned rules
Dashboard	ASP.NET Core Razor Pages with HTML and CSS	Authenticated inventory and findings views

Database	SQLite with Entity Framework Core	Small-scale persistent relational storage
Reports	HTML templates and CSV or JSON serializers	Human-readable reports and structured export
Development	Visual Studio and Git	Implementation and source version control

Table 2: Tools and Technologies

10.3 System Design / Architecture

An IT administrator enrolls an endpoint and provides its scoped submission credential. The administrator launches the agent and starts collection. The agent creates a timestamped JSON snapshot and submits it over HTTPS. The API authenticates the endpoint, validates the payload, applies the selected rule version, and stores inventory and results. An authenticated dashboard reads this central record and generates reports. Endpoint credentials permit submission for that endpoint only and do not grant dashboard access.

The core data flow is:
Windows evidence collection → validated snapshot submission → rule evaluation and prioritization → database persistence → dashboard and reports. If submission fails, the agent records the error and retains the export for retry. Snapshot identifiers will prevent the same audit from being duplicated during retries.

10.4 Implementation Plan

- **Collection module:** Separate collectors for each category, command timeouts, structured parsing, and explicit evidence availability. Determine the original interactive user separately from the elevated process account.
- **Rule engine:** Deterministic predicates with expected values, applicability conditions, severity, evidence references, and remediation text. Preserve every audit rule version.
- **Prioritization module:** Generate per-finding severity and an endpoint score. Show incomplete coverage and avoid silently converting errors into passes.
- **Dashboard module:** Login, endpoint list, inventory details, last audit timestamp, historical results, finding filters, and report download.
- **Reporting module:** Include endpoint identity, collection time, rule version, score, coverage, finding evidence, and remediation. Escape endpoint-supplied strings before producing HTML.
- **Security module:** Separate endpoint submission credentials and dashboard authentication, validate input, hash dashboard passwords, restrict report access, and avoid logging credentials.

10.5 Testing and Evaluation

Collectors will be checked against independently inspected Windows settings and manually recorded inventory values. Each of the 10 rules will have at least four controlled cases covering its applicable outcomes or evidence failures, producing at least 40 expected-result cases. Tests will verify that missing data never produces PASS. Collection failures, malformed payloads, unavailable APIs, unsupported antivirus providers, duplicate submissions, and unauthorized access will also be tested.

End-to-end trials will use at least three Windows endpoints or virtual machines and three audit repetitions per endpoint. Dangerous test changes will be restricted to isolated virtual machines with snapshots; production protections will not be disabled. The evaluator will compare exact rule outcomes and inventory accuracy against a manual reference record, report agreement and errors, and verify that every report is linked to the correct endpoint.

The manual and automated workflows will evaluate the same checklist and produce equivalent report content. Measure elapsed time from starting inspection to completing the report, separately recording operator effort and automated collection time. Report median times, variability, and percentage time reduction. A target of at least 30% lower median completion time will guide evaluation; it is a target rather than a claimed result.

A small usability evaluation will recruit approximately three to five consenting participants with IT knowledge where available. Tasks will include locating an endpoint, interpreting a finding, checking its evidence, and exporting a report. Record task completion, feedback, and a five-point ease-of-use rating. Small participant numbers and the controlled environment will be stated as limitations.

11.0 Expected Deliverables

- A portable Windows endpoint audit executable and documented collection interface.
- A documented catalogue of at least 10 rules, applicability conditions, severity assignments, and recommendations.
- A centralized API, authenticated dashboard, and database supporting at least three endpoints.
- HTML audit reports and CSV or JSON exports with evidence, timestamps, score, and coverage.
- Source code, deployment instructions, user guide, database design, and system architecture documentation.
- A test dataset using synthetic or consented evidence, evaluation results, final project report, and demonstration materials.

12.0 Risk Plan

Each risk is assessed using a 3 × 3 Probability and Impact Matrix, which combines the likelihood of a risk occurring with its effect on the project objectives to give an overall risk level. The rating definitions are given in Table 3, the risk level is read from the matrix in Table 4, and the resulting risk register is shown in Table 5.

Rating	Probability (likelihood of occurring)	Impact (effect on the project if it occurs)
High	Likely to occur during the project (more than 60%).	Prevents an objective or a key deliverable from being achieved.
Medium	May occur during the project (30% to 60%).	Delays a milestone or reduces the quality of one module, but the objectives remain achievable.
Low	Unlikely to occur during the project (less than 30%).	Minor effect that can be absorbed within the current sprint.

Table 3: Probability and Impact Rating Definitions

Probability & Impact		Impact		
		Low	Medium	High
Probability	High	Low	Medium	High
	Medium	Low	Medium	Medium
	Low	Low	Low	Low

Table 4: 3 × 3 Probability and Impact Matrix

Risk	Consequence	Probability	Impact	Risk Level	Response
Dashboard expansion exceeds time	Incomplete integrated system	High	High	High	Keep on-demand audits and one administrator role; defer scheduling, alerts, and remote management.

Permissions or Windows edition differences	Incomplete collection	High	Medium	Medium	Detect permissions and unsupported features; use UNKNOWN or justified non-applicability; test multiple configurations.
Third-party antivirus inconsistencies	Misleading protection results	Medium	High	Medium	Separate registration from health; support documented providers and disclose unsupported status.
Rules misclassify legitimate settings	Unnecessary remediation	Medium	Medium	Medium	Review baseline with supervisor; document exceptions and use WARNING for policy-dependent checks.
Small or homogeneous test sample	Limited generalization	High	Medium	Medium	Use varied configurations and state sample limitations in the final report.
Sensitive inventory is	Loss of confidentiality	Low	High	Low	Use consented lab

exposed					devices, authenticated access, HTTPS, scoped credentials, and minimal collection.
Network submission fails	Missing centralized record	Medium	Low	Low	Keep local export, show failure clearly, and support validated idempotent retry.
Unsafe test configuration changes	Reduced test device protection	Low	Medium	Low	Use isolated VMs and restore snapshots after controlled testing.

Table 5: Risk Register with Probability, Impact, and Risk Level

Risks are listed from the highest to the lowest risk level. Medium-level risks will be monitored and addressed through appropriate mitigation measures during development, while low-level risks will be managed as necessary. The risk register will be reviewed with the supervisor at the end of each sprint, and the probability, impact, and risk level of each risk will be updated when circumstances change.

13.0 Proposed Gantt chart

No.	Task Title	Start Date	End Date	Duration (Days)	% of Task Completed
1	**Phase 1 - Project Initiation**	**07/09/26**	**18/10/26**	**42**	**100%**
1.1	Project Preparation & Idea Exploration	07/09/26	20/09/26	14	100%
1.2	Requirement Gathering & Problem Analysis	14/09/26	11/10/26	28	100%
1.3	Technology Stack, Development Environment & GitHub Repository Setup	21/09/26	04/10/26	14	100%
1.4	Early Prototype: Simple UI & Backend	21/09/26	18/10/26	28	100%
1.5	Literature Review & Existing System Study	05/10/26	18/10/26	14	100%
2	**Phase 2 - System Design**	**19/10/26**	**15/11/26**	**28**	**32%**
2.1	Architecture, Database (ERD) & UML Design	19/10/26	01/11/26	14	64%
2.2	Audit Rule Catalogue (10 Rules) & Severity Design	02/11/26	15/11/26	14	0%
3	**Phase 3 - Development Sprints**	**02/11/26**	**07/02/27**	**98**	**0%**
3.1	Sprint 1: Collectors - Specs, Apps, Users, Accounts	02/11/26	15/11/26	14	0%
3.2	Sprint 2: Collectors - Antivirus, Firewall, Config	16/11/26	29/11/26	14	0%
3.3	Sprint 3: Rule Engine, Severity & Recommendation	30/11/26	13/12/26	14	0%
3.4	Sprint 4: SQLite	14/12/26	27/12/26	14	0%
3.5	Sprint 5: Dashboard - UI, History	28/12/26	10/01/27	14	0%
3.6	Sprint 6: Reports & Security Hardening	11/01/27	24/01/27	14	0%
3.7	Sprint 7: Integration, Bug Fixing & UI Polish	25/01/27	07/02/27	14	0%
4	**Phase 4 - Testing & Evaluation**	**08/02/27**	**14/03/27**	**35**	**0%**
4.1	Unit Testing	08/02/27	14/02/27	7	0%
4.2	System Testing on 3 Endpoints / VMs	15/02/27	21/02/27	7	0%
4.3	Manual vs Automated Audit Time Comparison	22/02/27	28/02/27	7	0%
4.4	Usability Testing	01/03/27	07/03/27	7	0%
4.5	Refinement Based on Test Feedback	08/03/27	14/03/27	7	0%
5	**Phase 5: Showcase & Final Submission**	**11/01/27**	**06/04/27**	**86**	**0%**
5.1	Final Report Writing	11/01/27	29/03/27	78	0%
5.2	Poster Preparation	15/02/27	21/02/27	7	0%
5.3	5-minute Video Demo	22/02/27	28/02/27	7	0%
5.4	Project ePortfolio	01/03/27	31/03/27	31	0%
5.5	Viva Presentation Preparation	01/04/27	05/04/27	5	0%
5.6	Viva	06/04/27	06/04/27	1	0%

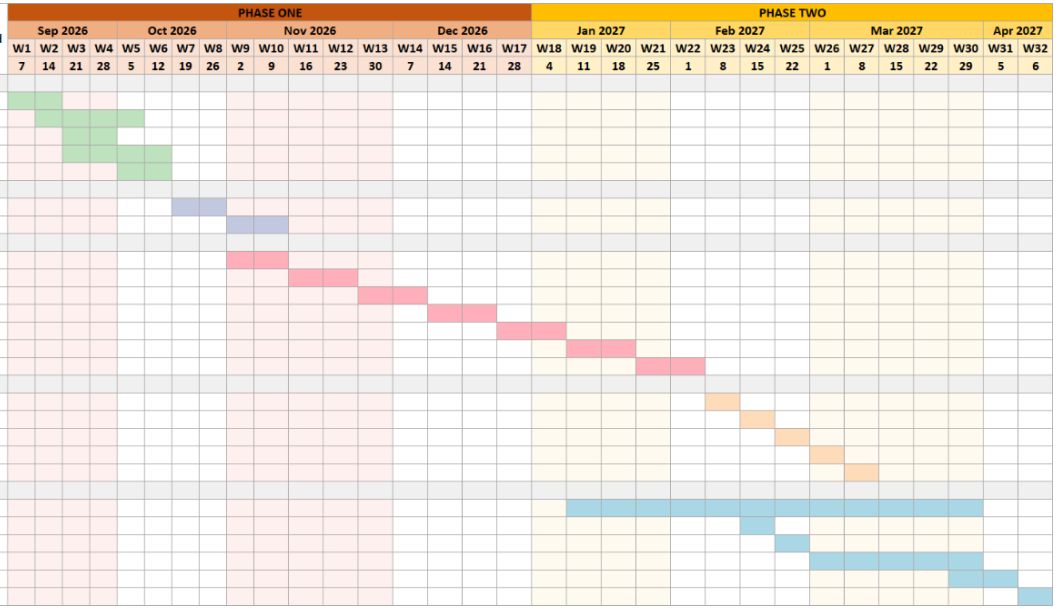


Figure 1: Project Gantt Chart (September 2026 - April 2027)

14.0 Resources Required

Hardware resources include a development laptop capable of running Visual Studio and virtual machines, access to at least three Windows endpoint instances, and a local network or controlled server environment. Physical devices and virtual machines may be combined. A server may run on the development computer for the demonstration, provided resource constraints are measured.

Software resources include appropriately licensed Windows test installations, a supported .NET SDK, Visual Studio, PowerShell, SQLite, Git, and a web browser. An HTTPS certificate trusted by the test devices will be required; certificate validation will remain enabled. No paid endpoint security platform or cloud service is required for the proposed local demonstration.

Test data will comprise synthetic inventory records, controlled security configuration states, and consented endpoint snapshots. Access to an IT practitioner or the supervisor is required to review rule assumptions. Participants will be informed about collected fields, and identifiable data will be removed from published examples. Exact costs will depend on existing university equipment and software access.

15.0 Expected Outcome and Significance

The expected outcome is an integrated prototype that consistently collects selected Windows endpoint evidence and converts it into traceable audit findings and reports. IT personnel should be able to compare several endpoints, identify missing information, and prioritize configuration weaknesses without repeating the entire collection and recording process manually.

The cybersecurity contribution is the documented connection between observed security controls, baseline rules, severity decisions, and remediation guidance. Accuracy and completeness matter more than the number of dashboard features. The project will demonstrate how uncertainty and policy-dependent findings can be presented without falsely declaring an endpoint secure.

The project may reduce audit effort for small environments and improve the consistency of reports. These benefits will be supported or qualified by the evaluation results. Its findings will apply to the tested Windows configurations and selected rules; a successful demonstration will not establish enterprise scalability, complete organizational compliance, or protection from every threat.

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